

Bootstrap:

Bootstrap is an open-source and free front-end framework that helps in creating mobile-first and responsive websites efficiently. It offers a wide range of ready-to-use CSS classes, a flexible grid system, and built-in UI elements such as buttons, forms, modals, and navigation bars. With Bootstrap, developers can design modern and uniform web layouts without having to write extensive CSS code manually.

There are many ways to work with Bootstrap :

1 Using CDN :

CDN (Content Delivery Network) :

A Content Delivery Network (CDN) is a group of servers distributed across different locations worldwide that store and deliver web content efficiently to users depending on their geographical area.

In the case of Bootstrap, a CDN allows developers to access Bootstrap's CSS and JavaScript files directly from online servers, removing the need to download or host the files locally.

This is the **quickest** method and is recommended for simple projects or when you only need to include the compiled CSS and JavaScript.

You link directly to Bootstrap's files hosted on a CDN (like jsDelivr) in your HTML file's `<head>` for CSS and before the closing `</body>` tag for JavaScript. This requires an **internet connection**.

```
<!-- Bootstrap CSS via CDN -->
```

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
```

```
<!-- Bootstrap JS via CDN -->
```

```
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></scri
pt>
```

Example:

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <!-- Bootstrap CSS via CDN -->
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

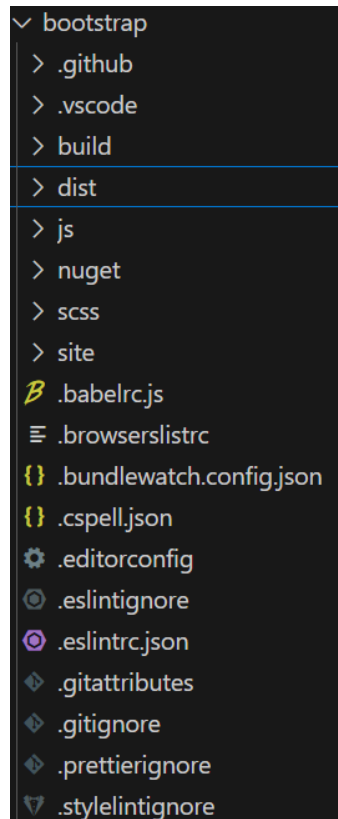
  <!-- Bootstrap JS via CDN -->
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.
js"></script>
</head>
<body>
  <button class="btn btn-primary">Click Me</button>
</body>
</html>
```

2. Downloading Bootstrap Files

Find github link from this documentation and clone it

<https://getbootstrap.com/docs/5.3/getting-started/contents/#css-files>

After cloning it will look like this



Link file to our html code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <link rel="stylesheet" href="./bootstrap/dist/css/bootstrap.min.css">
</head>
<body>
  <button class="btn btn-secondary">Click ME</button>
</body>
</html>
```

3. Using NPM :

If you are working with Node.js or modern web tools, install via npm:

Run command : `npm install bootstrap@5.3.8`

Add this to our code

```
<link rel="stylesheet"
href="./node_modules/bootstrap/dist/css/bootstrap.min.css">
```

Bootstrap basics :

Feature	Description
Responsive Grid System	Uses a flexible 12-column layout that adjusts automatically to different screen sizes.
Pre-styled Components	Includes built-in components like buttons, alerts, cards, and navbars.
Utility Classes	Offers quick CSS helpers for margin, padding, colors, text alignment, etc.
Mobile-First Design	Ensures layouts are optimized for mobile devices first, then scale up for larger screens.
Customizable	Supports Sass variables and themes for easy customization.
Cross-Browser Compatibility	Works well across modern browsers like Chrome, Edge, Safari, and Firefox.

Bootstrap Spacing System :

Bootstrap uses **utility classes** to manage **spacing** easily — without writing extra CSS.

Syntax : `{property}{sides}-{size}`

1. Property

- `m` → margin
- `p` → padding

2. Side (optional)

You can target specific sides:

Shortcut	Meaning
<code>t</code>	top
<code>b</code>	bottom
<code>s</code>	start (left in LTR, right in RTL)
<code>e</code>	end (right in LTR, left in RTL)
<code>x</code>	left and right
<code>y</code>	top and bottom

Example:

- `mt-3` → margin top = 1rem (approx 16px)
- `mb-5` → margin bottom = 3rem
- `px-4` → padding left and right = 1.5rem
- `py-2` → padding top and bottom = 0.5rem

3. Size scale

Size	Spacing
0	0
1	0.25rem (≈ 4px)
2	0.5rem (≈ 8px)
3	1rem (≈ 16px)
4	1.5rem (≈ 24px)
5	3rem (≈ 48px)
auto	automatic margin (used for centering)

Bootstrap CSS

No.	Topic
1. Containers	Containers are the main layout wrapper in Bootstrap that center your content and provide responsive padding. (<code>.container</code> or <code>.container-fluid</code>)
2. Grid System	Bootstrap uses a 12-column grid for responsive layouts, allowing you to easily create rows and columns using <code>.row</code> and <code>.col-*</code> classes.
3. Margin (m)	Adds space outside an element. Classes like <code>m-3</code> , <code>mt-4</code> , <code>mb-2</code> manage margin sizes.
4. Padding (p)	Adds space inside an element. Classes like <code>p-4</code> , <code>pt-3</code> , <code>pb-2</code> control internal spacing.

5. Background Colors	Classes like <code>bg-primary</code> , <code>bg-dark</code> , <code>bg-warning</code> provide quick background styling.
6. Text Colors	Change text color using classes like <code>text-danger</code> , <code>text-success</code> , or <code>text-muted</code> .
7. Typography (Fonts)	Control text size, weight, and style using <code>fw-bold</code> , <code>fst-italic</code> , <code>lh-lg</code> , and heading classes like <code>h1–h6</code> .
8. Buttons	Predefined button styles such as <code>btn-primary</code> , <code>btn-success</code> , and <code>btn-outline-*</code> provide consistent, responsive buttons.
9. Images	Make images responsive and styled using <code>img-fluid</code> , <code>rounded</code> , and <code>img-thumbnail</code> .
10. Inputs & Forms	The <code>.form-control</code> class gives a clean, modern style to text fields, checkboxes, and dropdowns.
11. Alerts	Display notifications using <code>.alert</code> with contextual classes like <code>alert-warning</code> or <code>alert-success</code> .
12. Badges	Small count indicators or labels used inside buttons or headings with <code>.badge</code> and <code>bg-*</code> .
13. Cards	Ready-made flexible content containers with options for images, text, and buttons (<code>.card</code> , <code>.card-body</code>).
14. Navbar	Responsive navigation bar that collapses on small devices (<code>.navbar</code> , <code>.navbar-expand-lg</code> , <code>.navbar-light</code>).
15. Tables	Responsive, styled tables using <code>.table</code> , <code>.table-striped</code> , and <code>.table-hover</code> .
16. Icons	Use Bootstrap Icons (via CDN) for scalable and easy-to-style vector icons.

17. List Groups	Create neatly styled lists using <code>.list-group</code> and <code>.list-group-item</code> .
18. Progress Bars	Show task progress with <code>.progress</code> and <code>.progress-bar</code> classes.
19. Shadows and Borders	Add effects like <code>shadow</code> , <code>shadow-lg</code> , and <code>border</code> , <code>rounded</code> , <code>border-0</code> for smooth styling.
20. Utilities (Display, Flex, Align)	Utility classes like <code>d-flex</code> , <code>justify-content-center</code> , and <code>align-items-center</code> make layout and positioning simple.

Flex :

Flexbox utilities make it easier to design flexible, responsive, and well-aligned layouts without writing custom CSS.

These classes apply CSS Flexbox properties directly to your HTML elements, letting you control positioning, alignment, and spacing quickly

Category	Classes	Purpose
Display	<code>.d-flex</code>	Converts an element into a flex container
Direction	<code>.flex-row</code> , <code>.flex-column</code>	Sets layout direction horizontally or vertically
Wrapping	<code>.flex-wrap</code> , <code>.flex-nowrap</code>	Controls whether items wrap to the next line
Horizontal alignment	<code>.justify-content-start</code> , <code>.justify-content-center</code> , <code>.justify-content-end</code> ,	Positions items along the main axis

	<code>.justify-content-between,</code> <code>.justify-content-around</code>	
Vertical alignment	<code>.align-items-start,</code> <code>.align-items-center,</code> <code>.align-items-end</code>	Aligns items along the cross-axis
Self alignment	<code>.align-self-start,</code> <code>.align-self-center,</code> <code>.align-self-end</code>	Aligns individual flex items
Ordering	<code>.order-1, .order-2, etc.</code>	Changes the visual order of items
Flex growth	<code>.flex-grow-1, .flex-shrink-0</code>	Controls how items expand or shrink
Responsive display	<code>.d-sm-flex, .d-md-flex,</code> <code>.d-lg-flex</code>	Enables flexbox only on specific breakpoints

Example :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Scalability Example</title>
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="p-5">
  <div class="container">
    <h3 class="text-success mb-4 text-center">Scalable Product Grid</h3>
    <div class="row g-4">
      <div class="col-12 col-sm-6 col-md-4 col-lg-3">
        <div class="card h-100 text-center">
          
          <div class="card-body">
            <h5 class="card-title">Product 1</h5>
            <p class="card-text">Responsive and scalable layout
```

```

using grid.</p>
    </div>
  </div>
</div>

  <div class="col-12 col-sm-6 col-md-4 col-lg-3">
    <div class="card h-100 text-center">
      
      <div class="card-body">
        <h5 class="card-title">Product 2</h5>
        <p class="card-text">Adapts smoothly to screen size
changes.</p>
      </div>
    </div>
  </div>

  <div class="col-12 col-sm-6 col-md-4 col-lg-3">
    <div class="card h-100 text-center">
      
      <div class="card-body">
        <h5 class="card-title">Product 3</h5>
        <p class="card-text">Fully scalable layout using
Bootstrap grid.</p>
      </div>
    </div>
  </div>
</div>
</div>
</body>
</html>

```

Grid :

The **Grid System** is the foundation of Bootstrap's layout and responsiveness. It's a way to arrange content in rows and columns that automatically adjust based on screen size. Bootstrap's grid is built with **12 columns** using **Flexbox**.

a. Containers

All content must be wrapped in a container to align and pad the content correctly.

- **.container**: Provides a **responsive fixed width** container. Its width changes at each of Bootstrap's predefined breakpoints.
- **.container-fluid**: Provides a **full-width** container that spans the entire width of the viewport on all devices.

b. Rows and Columns

- **Rows (.row)** must be placed inside a container and create horizontal groups of columns.
- **Columns (.col-*)** are the direct children of rows and hold your content. The grid is based on a **12-column system**. You define how many of those 12 columns a content section should span.

C. Breakpoints (Grid Tiers)

Bootstrap uses six default breakpoints to control how your layout behaves across different screen sizes. You use these prefixes in your column classes.

Breakpoint	Class Prefix	Viewport Size	Use Case
X-Small	None (e.g., .col-4)	< 576px	Portrait phones
Small	.col-sm-	>=576px	Landscape phones
Medium	.col-md-	>=768px	Tablets

Large	.col-lg-	>=992px	Desktops
X-Large	.col-xl-	>=1200px	Large desktops
XX-Large	.col-xxl-	>=1400px	Ultra-large desktops

Example:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Grid Basic</title>
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body>
  <div class="container mt-5">
    <h2 class="text-center text-primary mb-4">Bootstrap Grid Example</h2>

    <div class="row">
      <div class="col-12 col-md-4 bg-success text-white p-3 border
border-black">Column 1</div>
      <div class="col-12 col-md-4 bg-warning text-white p-3 border
border-black">Column 2</div>
      <div class="col-12 col-md-4 bg-danger text-white p-3 border
border-black">Column 3</div>

      <div class="col-12 col-md-4 bg-success text-white p-3 border
border-black">Column 1</div>
      <div class="col-12 col-md-4 bg-warning text-white p-3 border
border-black">Column 2</div>
      <div class="col-12 col-md-4 bg-danger text-white p-3 border
border-black">Column 3</div>
```

```
<div class="col-12 col-md-4 bg-success text-white p-3 border
border-black">Column 1</div>
<div class="col-12 col-md-4 bg-warning text-white p-3 border
border-black">Column 2</div>
<div class="col-12 col-md-4 bg-danger text-white p-3 border
border-black">Column 3</div>
</div>
</div>
</body>
</html>
```

Themes:

A **Bootstrap theme** is a **customized design layer** built on top of the Bootstrap framework that modifies its **default visual style**, such as colors, typography, spacing, and component appearances, while preserving the **core layout system and responsive behavior** provided by Bootstrap.

In other words, a theme allows developers and designers to give their websites a **unique brand identity and aesthetic** without needing to rebuild the front-end structure from scratch.

Important Characteristics of Bootstrap Themes:

1. Built on Bootstrap's Core

Creating a theme or custom design that still relies on Bootstrap's fundamental structure such as its grid layout, components, and utility classes while only altering the visual presentation (like color palette, typography, or button styles).

Even though the theme looks different, the core Bootstrap framework continues to handle layout, spacing, and responsiveness.

Example :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Core Example</title>
  <link rel="stylesheet"
href="./node_modules/bootstrap/dist/css/bootstrap.min.css">
</head>
<body>

  <div class="container text-center my-5">
    <h1 class="text-primary">Built on Bootstrap's Core</h1>
    <p class="lead">This layout uses only Bootstrap's grid and utility
classes.</p>

    <div class="row mt-4">
      <div class="col-md-4">
        <div class="card">
          <div class="card-body">
            <h5 class="card-title">Column 1</h5>
            <p class="card-text">Uses Bootstrap's default card component.</p>
            <button class="btn btn-primary">Click</button>
          </div>
        </div>
      </div>

      <div class="col-md-4">
        <div class="card">
          <div class="card-body">
            <h5 class="card-title">Column 2</h5>
            <p class="card-text">Responsive with Bootstrap grid.</p>
            <button class="btn btn-success">Click</button>
          </div>
        </div>
      </div>

      <div class="col-md-4">
        <div class="card">
          <div class="card-body">
            <h5 class="card-title">Column 3</h5>
            <p class="card-text">No custom CSS used here.</p>
            <button class="btn btn-danger">Click</button>
          </div>
        </div>
      </div>
    </div>
  </div>
</body>
</html>
```

```
        </div>
      </div>
    </div>
  </div>
</div>

<script
src="./node_modules/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

2. Customization Through Sass or CSS Variables

Example

1 Make Sure bootstrap framework is present on your system if not then use this command

npm install bootstrap@5.3.8

2 Make your SASS also present inside your system, if not follow this command

npm install bootstrap sass

3 Now generate a css file from scss file, using command

npx sass scss/custom.scss css/custom.css

Npx : **npx** is a Node.js tool that runs packages (like Sass) directly without global installation, meaning it executes the command temporarily — “run this package just once, right now.”

Sass : **sass** is the **Sass compiler** — a program that converts **.scss** (Sass files) into regular **.css** files that browsers can understand.

scss/custom.scss : input folder / input file
css/custom.css : output folder / output file

Folder Structure look like this :

bootstrap-sass-custom/

├── node_modules/

├── scss/

| └── custom.scss

├── css/

| └── custom.css

├── index.html

└── package.json

Now Link file with custom.css

Note : Ignore all the Warning just start live server

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <link rel="stylesheet" href="./css/style.css">
</head>
<body>
  <div class="p-5 text-center">
    <h1 class="text-primary">My Coral Theme</h1>
    <p class="text-secondary">This is a customized Bootstrap theme.</p>
    <button class="btn btn-primary">Try Now</button>
  </div>
</body>
</html>
```


3. Consistency Across Components

One of Bootstrap's strongest advantages is that all its UI elements (buttons, forms, alerts, modals, navbars, etc.) are built using a consistent design system — shared colors, font sizes, spacing, and border styles.

This means every component feels visually connected, creating a unified look across the entire website or app.

Even when you customize a theme (like changing the color palette), the changes automatically apply to all Bootstrap components — maintaining harmony.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width,initial-scale=1" />
  <title>Theme override test</title>
  <link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&display=swap" rel="stylesheet">
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    :root{
      --bs-primary: #e76f51;          /* coral */
      --bs-success: #2a9d8f;         /* teal */
      --bs-font-sans-serif: 'Poppins', system-ui, -apple-system, "Segoe UI",
Roboto, "Helvetica Neue", Arial;
      --bs-body-bg: #fffdf9;
      --bs-body-color: #2b2b2b;
    }

    body {
      font-family: var(--bs-font-sans-serif);
      background-color: var(--bs-body-bg);
      color: var(--bs-body-color);
    }

    .btn-primary {
      background-color: var(--bs-primary) !important;
      border-color: var(--bs-primary) !important;
```

```

    }
  </style>
</head>
<body class="p-4">
  <div class="container text-center">
    <h2 class="text-primary mb-4">Custom theme override (test)</h2>
    <p class="lead">Primary should be coral, success should be teal, font =
Poppins.</p>

    <button class="btn btn-primary m-2">Primary Button</button>
    <button class="btn btn-success m-2">Success Button</button>

    <div class="alert alert-primary mt-4" role="alert">
      This is a themed alert (primary).
    </div>

    <div class="card mx-auto mt-4 shadow-sm" style="width: 18rem;">
      <div class="card-body">
        <h5 class="card-title text-primary">Themed Card</h5>
        <p class="card-text">Buttons, alert and headings should follow the
custom color.</p>
        <a href="#" class="btn btn-primary">Learn More</a>
      </div>
    </div>
  </div>
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.
js"></script>
</body>
</html>

```

4. Reusability and Scalability

Reusability in Bootstrap means creating consistent UI elements using Bootstrap's predefined classes and components that can be used multiple times throughout a project without writing custom CSS every time.

It allows developers to maintain a uniform design and reduce repetitive code.

Example :

```
<!DOCTYPE html>
```

```

<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width,initial-scale=1" />
  <title>Theme override test</title>
  <link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&display=swa
p" rel="stylesheet">
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    :root{
      --bs-primary: #e76f51;          /* coral */
      --bs-success: #2a9d8f;         /* teal */
      --bs-font-sans-serif: 'Poppins', system-ui, -apple-system, "Segoe UI",
Roboto, "Helvetica Neue", Arial;
      --bs-body-bg: #fffdf9;
      --bs-body-color: #2b2b2b;
    }

    body {
      font-family: var(--bs-font-sans-serif);
      background-color: var(--bs-body-bg);
      color: var(--bs-body-color);
    }

    .btn-primary {
      background-color: var(--bs-primary) !important;
      border-color: var(--bs-primary) !important;
    }
  </style>
</head>
<body class="p-4">
  <div class="container text-center">
    <h2 class="text-primary mb-4">Custom theme override (test)</h2>
    <p class="lead">Primary should be coral, success should be teal, font =
Poppins.</p>

    <button class="btn btn-primary m-2">Primary Button</button>
    <button class="btn btn-success m-2">Success Button</button>

    <div class="alert alert-primary mt-4" role="alert">
      This is a themed alert (primary).
    </div>

```

```

<div class="card mx-auto mt-4 shadow-sm" style="width: 18rem;">
  <div class="card-body">
    <h5 class="card-title text-primary">Themed Card</h5>
    <p class="card-text">Buttons, alert and headings should follow the
custom color.</p>
    <a href="#" class="btn btn-primary">Learn More</a>
  </div>
</div>
</div>
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.
js"></script>
</body>
</html>

```

Scalability in Bootstrap refers to designing a layout that **adapts automatically** to different screen sizes or future content changes.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Reusability Example</title>
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="p-5">

  <div class="container text-center">
    <h3 class="mb-4 text-primary">Reusable Bootstrap Components</h3>

    <!-- Reusable Buttons -->
    <button class="btn btn-primary m-2">Submit</button>
    <button class="btn btn-success m-2">Save</button>
    <button class="btn btn-danger m-2">Delete</button>

    <!-- Reusable Card -->
    <div class="card mt-4 mx-auto" style="width: 18rem;">
      
      <div class="card-body">

```

```

    <h5 class="card-title">Reusable Card</h5>
    <p class="card-text">This card uses Bootstrap's predefined structure
and classes.</p>
    <a href="#" class="btn btn-primary">Read More</a>
  </div>
</div>
</div>

</body>
</html>

```

5. Dark and Light Modes

Dark mode is a design feature that switches a website's or application's color scheme from light backgrounds with dark text to dark backgrounds with light text. It reduces glare, improves readability in low-light settings, and provides an elegant, modern look.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <link rel="stylesheet"
href="./node_modules/bootstrap/dist/css/bootstrap.min.css">
</head>
<body data-bs-theme="dark">
  <button class="btn btn-primary">Dark Mode Button</button>
</body>
</html>

```

6. Third-Party or Official Themes

Third-party or official Bootstrap themes are professionally designed style packages built on top of the Bootstrap framework.

They enhance or customize the default Bootstrap appearance by changing colors, typography, spacing, and component styles, while still using the same underlying Bootstrap structure and class names.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootswatch Cosmo Example</title>
  <!-- Bootswatch Cosmo Theme -->
  <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootswatch@5.3.3/dist/cosmo/bootstrap.min.cs
s">
</head>
<body class="p-4">

  <div class="container text-center">
    <h1 class="mb-4">Bootswatch: Cosmo Theme</h1>
    <button class="btn btn-primary m-2">Primary Button</button>
    <button class="btn btn-success m-2">Success Button</button>
    <button class="btn btn-danger m-2">Danger Button</button>
    <p class="mt-4">This is the <strong>Cosmo</strong> theme -- with flat
buttons and clean design.</p>
  </div>
</body>
</html>
```

Alert:

The **Alert** component is used to provide contextual feedback messages to the user (like success, warning, or error messages).

Example1 :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Alert Example</title>
  <link rel="stylesheet"
```

```

href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css">
</head>
<body class="p-4">

  <div class="container">
    <h2 class="mb-4">Bootstrap Alerts</h2>

    <div class="alert alert-primary" role="alert">
      This is a primary alert -- check it out!
    </div>

    <div class="alert alert-success" role="alert">
      Success! Your action was completed successfully.
    </div>

    <div class="alert alert-warning" role="alert">
      Warning! Please double-check your input.
    </div>

    <div class="alert alert-danger" role="alert">
      Error! Something went wrong.
    </div>

  </div>

</body>
</html>

```

Example2:

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Form Alert Example</title>
  <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css">
</head>
<body class="p-5">

  <div class="container">

```

```

<h2 class="mb-4 text-center">Contact Form</h2>

<!-- Alert (hidden initially) -->
<div id="successAlert" class="alert alert-success d-none" role="alert">
  Form Submitted Successfully
</div>

<!-- Form -->
<form id="contactForm">
  <div class="mb-3">
    <label for="name" class="form-label">Name</label>
    <input type="text" id="name" class="form-control" placeholder="Enter
your name" required>
  </div>

  <div class="mb-3">
    <label for="email" class="form-label">Email</label>
    <input type="email" id="email" class="form-control" placeholder="Enter
your email" required>
  </div>

  <button type="submit" class="btn btn-primary w-100">Submit</button>
</form>
</div>

<script>
  // Show alert when form is submitted
  document.getElementById('contactForm').addEventListener('submit',
function(event) {
    event.preventDefault(); // Prevent page reload
    document.getElementById('successAlert').classList.remove('d-none');
  });
</script>

</body>
</html>

```

Well:

The `.well` class was used to easily create an **inset effect** on an element's content. It added **padding**, a **light gray background color**, a subtle **border**, and **rounded corners** to make the content appear "sunken" into the page.

The Bootstrap `.well` class is **deprecated** in modern versions of Bootstrap (v4 and v5) but was a common component in **Bootstrap 3**.


```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Bootstrap 3 Well Example</title>
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
</head>
<body>

<div class="container">
  <header class="page-header">
    <h1>Bootstrap 3 <code>.well</code> Examples (Deprecated)</h1>
  </header>

  <h3>Default Well</h3>
  <div class="well">
    This is a default Bootstrap 3 Well. It gives the content a light gray,
    padded, and sunken appearance.
  </div>

  <h3>Large Well</h3>
  <div class="well well-lg">
    This is a Large Well (<code>.well-lg</code>) with increased padding.
  </div>

  <h3>Small Well</h3>
  <div class="well well-sm">
    This is a Small Well (<code>.well-sm</code>) with reduced padding.
  </div>

</div>

</body>
</html>

```

Badge:

A **Badge** in Bootstrap is a small, highlighted element used to display **notification counts**, **status labels**, or **emphasized information** next to text, links, or buttons.

It usually appears as a **pill-shaped or rounded indicator** with a colored background and light text — often used to grab attention to something important like “New”, “4”, or “Active”.

Type	Class/Feature	Short Description
Default Badge	<code>.badge</code>	Basic rectangular badge used for labeling or showing numbers.
Contextual Badge	<code>.bg-primary</code> , <code>.bg-success</code> , <code>.bg-danger</code> , etc.	Badges with different background colors to represent meaning (info, success, danger, etc.).
Pill Badge	<code>.rounded-pill</code>	Gives the badge a smooth, capsule-like (rounded) shape.
Button Badge	Badge inside a button	Displays notification counts or updates on a button.
Link Badge	Badge inside a link tag <code><a></code>	Acts as a clickable badge, often used in navigation or menus.
Light/Dark Mode Badge	<code>text-dark bg-light</code> or <code>text-light</code> <code>bg-dark</code>	Adjusts badge color according to theme (light or dark mode).

```
<!DOCTYPE html>  
<html lang="en">
```

```

<head>
  <meta charset="UTF-8">
  <title>Bootstrap 3 Well Example</title>
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
</head>
<body>

<div class="container">
  <header class="page-header">
    <h1>Bootstrap 3 <code>.well</code> Examples (Deprecated)</h1>
  </header>

  <h3>Default Well</h3>
  <div class="well">
    This is a default Bootstrap 3 Well. It gives the content a light gray,
    padded, and sunken appearance.
  </div>

  <h3>Large Well</h3>
  <div class="well well-lg">
    This is a Large Well (<code>.well-lg</code>) with increased padding.
  </div>

  <h3>Small Well</h3>
  <div class="well well-sm">
    This is a Small Well (<code>.well-sm</code>) with reduced padding.
  </div>

</div>

</body>
</html>

```

Label:

A **label** in Bootstrap (used in older versions like **Bootstrap 3**) was a small text indicator used to highlight or categorize content — such as status, tags, or keywords.

It looked similar to a **badge**, with a colored background and white text.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Labels Example</title>
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
</head>
<body>
  <div class="container">
    <h3>Product Status:</h3>
    <span class="label label-default">Default</span>
    <span class="label label-primary">Primary</span>
    <span class="label label-success">Success</span>
    <span class="label label-info">Info</span>
    <span class="label label-warning">Warning</span>
    <span class="label label-danger">Danger</span>
  </div>
</body>
</html>

```

Panels:

A **panel** in Bootstrap (introduced in **Bootstrap 3**) is a **boxed container** used to group related content together with a border, some padding, and an optional heading or footer.

Panels were commonly used to organize information, such as displaying data, forms, or messages inside a bordered section.

Panels were **removed after Bootstrap 3** and **replaced by Cards** in **Bootstrap 4 and 5**, which are more flexible and powerful.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Panel Example</title>
  <link rel="stylesheet"

```

```

href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
</head>
<body>
  <div class="container">
    <h2>Bootstrap Panel Example</h2>

    <div class="panel panel-default">
      <div class="panel-heading">Panel Heading</div>
      <div class="panel-body">This is the content inside the panel body.</div>
      <div class="panel-footer">Panel Footer</div>
    </div>

  </div>
</body>
</html>

```

Example2.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap 3 Panel Dashboard Example</title>
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
</head>
<body>
  <div class="container">
    <h2 class="text-center">User Dashboard</h2>
    <div class="row">

      <!-- User Info Panel -->
      <div class="col-md-4">
        <div class="panel panel-primary">
          <div class="panel-heading">Profile Overview</div>
          <div class="panel-body">
            <p><strong>Name:</strong> Aakash Vishwakarma</p>
            <p><strong>Email:</strong> aakash@example.com</p>
            <p><strong>Role:</strong> Web Developer</p>
          </div>
          <div class="panel-footer text-right">
            <button class="btn btn-sm btn-default">Edit Profile</button>
          </div>
        </div>
      </div>
    </div>
  </div>

```

```

        </div>
    </div>
</div>

<!-- Recent Activity Panel -->
<div class="col-md-4">
    <div class="panel panel-info">
        <div class="panel-heading">Recent Activity</div>
        <div class="panel-body">
            <ul>
                <li>Uploaded a project to GitHub</li>
                <li>Updated Bootstrap tutorial</li>
                <li>Commented on a forum post</li>
            </ul>
        </div>
        <div class="panel-footer text-right">
            <a href="#" class="btn btn-sm btn-info">View All</a>
        </div>
    </div>
</div>

<!-- Notifications Panel -->
<div class="col-md-4">
    <div class="panel panel-warning">
        <div class="panel-heading">Notifications</div>
        <div class="panel-body">
            <p><span class="label label-danger">3</span> Pending Tasks</p>
            <p><span class="label label-success">2</span> Approved Requests</p>
            <p><span class="label label-info">1</span> New Message</p>
        </div>
        <div class="panel-footer text-right">
            <button class="btn btn-sm btn-warning">Check Now</button>
        </div>
    </div>
</div>

</div>
</div>
</body>
</html>

```

Pagination:

Pagination in Bootstrap is a **navigation component** that helps users move between multiple pages of related content — for example, navigating between different pages of a blog, product list, or table.

It provides a **structured and user-friendly way** to handle large datasets by dividing them into smaller, manageable sections.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap 3 Panel Dashboard Example</title>
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
</head>
<body>
  <div class="container">
    <h2 class="text-center">User Dashboard</h2>
    <div class="row">

      <!-- User Info Panel -->
      <div class="col-md-4">
        <div class="panel panel-primary">
          <div class="panel-heading">Profile Overview</div>
          <div class="panel-body">
            <p><strong>Name:</strong> Aakash Vishwakarma</p>
            <p><strong>Email:</strong> aakash@example.com</p>
            <p><strong>Role:</strong> Web Developer</p>
          </div>
          <div class="panel-footer text-right">
            <button class="btn btn-sm btn-default">Edit Profile</button>
          </div>
        </div>
      </div>

      <!-- Recent Activity Panel -->
      <div class="col-md-4">
        <div class="panel panel-info">
          <div class="panel-heading">Recent Activity</div>
          <div class="panel-body">
            <ul>
              <li>Uploaded a project to GitHub</li>
            </ul>
          </div>
        </div>
      </div>
    </div>
  </div>
```

```

        <li>Updated Bootstrap tutorial</li>
        <li>Commented on a forum post</li>
    </ul>
</div>
<div class="panel-footer text-right">
    <a href="#" class="btn btn-sm btn-info">View All</a>
</div>
</div>
</div>

<!-- Notifications Panel -->
<div class="col-md-4">
    <div class="panel panel-warning">
        <div class="panel-heading">Notifications</div>
        <div class="panel-body">
            <p><span class="label label-danger">3</span> Pending Tasks</p>
            <p><span class="label label-success">2</span> Approved Requests</p>
            <p><span class="label label-info">1</span> New Message</p>
        </div>
        <div class="panel-footer text-right">
            <button class="btn btn-sm btn-warning">Check Now</button>
        </div>
    </div>
</div>

</div>
</div>
</body>
</html>

```

Pager:

A *pager* in Bootstrap provides simple navigation between two pages or content sections, typically using “Previous” and “Next” links. It helps users move backward and forward in a series of related items such as articles, steps, or forms.

It was available in **Bootstrap 3** as `.pager`.

In **Bootstrap 4 and 5**, the **pager class was removed**, and replaced with the **pagination component**

Example:

```
<!DOCTYPE html>
```



```

<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <!-- <link rel="stylesheet"
href="./node_modules/bootstrap/dist/css/bootstrap.min.css"> -->
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
  <!-- <script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script
>
  <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/js/bootstrap.min.js"></scr
ipt> -->
</head>
<body>

<div class="container">
  <h2>Pager</h2>
  <p>The .pager class provides previous and next buttons (links):</p>
  <ul class="pager">
    <li><a href="#">Previous</a></li>
    <li><a href="#">Next</a></li>
  </ul>
</div>

</body>
</html>

```

Carousel :

The Bootstrap Carousel is a dynamic slideshow element that automatically cycles through content such as images or text with smooth transitions. It enhances user engagement and provides an interactive way to display multiple items in a limited space.

JS styling cdn/downloaded link must be present to work properly in carousel

Example 1 :

```

<!DOCTYPE html>
<html lang="en">

<head>

```

```

    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <link rel="stylesheet"
href="./node_modules/bootstrap/dist/css/bootstrap.min.css">
</head>

<body>
    <div id="carouselBasic" class="carousel slide" data-bs-ride="carousel">
        <div class="carousel-inner">
            <div class="carousel-item active">
                
                <div class="carousel-caption bg-dark bg-opacity-50 rounded">
                    <h5>Nature Beauty</h5>
                    <p>Experience the freshness of green landscapes.</p>
                </div>
            </div>
            <div class="carousel-item">
                
                <div class="carousel-caption bg-dark bg-opacity-50 rounded">
                    <h5>Urban Adventure</h5>
                    <p>Explore the modern life of bustling cities.</p>
                </div>
            </div>
        </div>

        <button class="carousel-control-prev" type="button"
data-bs-target="#carouselBasic" data-bs-slide="prev">
            <span class="carousel-control-prev-icon"></span>
        </button>
        <button class="carousel-control-next" type="button"
data-bs-target="#carouselBasic" data-bs-slide="next">
            <span class="carousel-control-next-icon"></span>
        </button>
    </div>
    <script
src="./node_modules/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
    <!-- <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.
js"></script> -->

</body>
</html>

```

Example2

Default delay time is 5000ms.

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <link rel="stylesheet"
href="./node_modules/bootstrap/dist/css/bootstrap.min.css">
</head>

<body>
  <div id="fadeCarousel" class="carousel slide carousel-fade"
data-bs-ride="carousel">
    <div class="carousel-inner">
      <div class="carousel-item active" data-bs-interval="1000">
        
      </div>
      <div class="carousel-item" data-bs-interval="3000">
        
      </div>
    </div>
  </div>
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.
js"></script>

</body>

</html>
```

Progress Bar :

A Bootstrap Progress Bar is a visual component that displays how much of a task or operation has been completed — such as uploading a file, loading content, or finishing a process.

It uses simple HTML and Bootstrap's built-in classes, allowing you to easily style and control progress visually.

Type	Class	Purpose
Basic	<code>.progress,</code> <code>.progress-bar</code>	Displays progress level
Colored	<code>.bg-success,</code> <code>.bg-warning</code> , etc.	Indicates status
Striped	<code>.progress-bar-striped</code>	Adds striped pattern
Animated	<code>.progress-bar-animated</code>	Adds animation to stripes
Stacked	Multiple <code>.progress-bar</code>	Shows multiple progress values

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Bootstrap Progress Bars - Full Example</title>
  <link rel="stylesheet"
href="./node_modules/bootstrap/dist/css/bootstrap.min.css">
</head>

<body class="p-4 bg-light">

  <div class="container">
    <h2 class="text-center mb-4">Bootstrap Progress Bars - All
Scenarios</h2>

    <!-- Basic Progress Bar -->
```

```

<h5>1. Basic Progress Bar</h5>
<div class="progress mb-4">
  <div class="progress-bar" style="width: 60%;">60%</div>
</div>

<!-- Colored Progress Bars -->
<h5>2. Contextual Colors</h5>
<div class="progress my-2">
  <div class="progress-bar bg-success" style="width: 80%;">Upload
  Success</div>
</div>
<div class="progress my-2">
  <div class="progress-bar bg-info" style="width:
  50%;">Processing</div>
</div>
<div class="progress my-2">
  <div class="progress-bar bg-warning" style="width:
  30%;">Pending</div>
</div>
<div class="progress my-2">
  <div class="progress-bar bg-danger" style="width:
  90%;">Error</div>
</div>

<!-- Striped Progress Bar -->
<h5 class="mt-4">3. Striped Progress Bar</h5>
<div class="progress mb-4">
  <div class="progress-bar progress-bar-striped bg-primary"
  style="width: 75%;">Uploading...</div>
</div>

<!-- Animated Progress Bar -->
<h5>4. Animated Progress Bar</h5>
<div class="progress mb-4">
  <div class="progress-bar progress-bar-striped
  progress-bar-animated bg-success" style="width: 65%;">Loading
  Data...</div>
</div>

```

```
<!-- Stacked Progress Bars -->
<h5>5. Stacked Progress Bar (Multiple Tasks)</h5>
<div class="progress mb-4">
  <div class="progress-bar bg-success" style="width: 40%">Task
1</div>
  <div class="progress-bar bg-warning" style="width: 25%">Task
2</div>
  <div class="progress-bar bg-danger" style="width: 15%">Task
3</div>
</div>

<!-- Example with Labels -->
<h5>6. Progress Bar with Labels and Tooltips</h5>
<div class="progress mb-4" style="height: 25px;">
  <div class="progress-bar bg-info progress-bar-striped
progress-bar-animated" style="width: 70%;" title="70% completed">
    70% Completed
  </div>
</div>
</div>

<script
src="./node_modules/bootstrap/dist/js/bootstrap.bundle.min.js"></scri
pt>
</body>

</html>
```